Gyandeep Mishra

Software Development & Machine Learning Engineer
Passionate programmer, scalable system designer, all-round tech enthusiast

Gmail.com | portfolio.com | github.com | linkedin.com

Academic Qualifications

Year	Degree	Institute	Performance
2022 – 2026	B.Tech (ECE)	SRM Institute of Science & Technology, Kattankulathur	CGPA: 8.46 / 10

Professional Experience

Technical Intern | *NeGD, Ministry of Electronics & IT (MeitY), New Delhi* | *Jul 2026 – Present*

- Developed an AI-powered Parliamentary Question & Answer Retrieval System using Agentic RAG, OpenSearch, vector embeddings, and hybrid search, enabling accurate semantic retrieval across multilingual parliamentary documents.
- Engineered document ingestion and parsing pipelines using Azure Document Intelligence for structured extraction of English and Hindi PDFs, while analyzing and optimizing a FastAPI-based backend to improve system scalability, maintainability, and performance.

Data Analytics Intern | *Deloitte, Remote* | *Jul 2025 – Aug 2025*

- Completed Deloitte Data Analytics Job Simulation with hands-on tasks in forensic data analysis and investigative technology.
- Applied business analytics to real-world case studies, gaining practical insights into enterprise consulting workflows.

Technical Skills

Programming Languages	Java, Python, C++, C
Areas of Interest	Artificial Intelligence, Agentic RAG, LangChain, Machine Learning, Deep Learning, Data Science, Natural Language Processing
Databases	MySQL, MongoDB, OpenSearch
Tools and Technologies	Jupyter Notebook, FastAPI, HTML, Excel, Azure Document Intelligence, Visual Studio Code, Azure OpenAI, Flask, Streamlit, MS Office, Git and GitHub, Vector Search, Azure AI Services

Key Projects

Multimodal Deep Learning Based Precision Treatment for Lung Cancer | *Python, TensorFlow, EfficientNetB0, OpenCV, Scikit-Learn, NumPy, Pandas*

- Built multimodal deep learning framework integrating CT imaging and clinical patient data for lung cancer subtype classification.
- Achieved 96.8% classification accuracy using feature-level fusion, outperforming individual CT (87.4%) and clinical (88.8%) models.

Soil Moisture and Fertility Analysis with Camera Integration | *Arduino, Yolov8, Flask, Kaggle Notebook, Python, HTML, CSS*

- Built IoT-enabled system with Arduino and YOLOv8 to analyze 500+ soil images daily.
- Achieved 92% fertility detection and 88% moisture classification, reducing water use by 20–25%.
- Automated data pipeline using Python scripts, reducing manual workload.

High-Performance Multi-threaded DPI Engine for Network Traffic Analysis | *C++, Multithreading, PCAP, TLS, Network Security*

- Designed and implemented a Deep Packet Inspection (DPI)-based network traffic analyzer, achieving 100 % packet processing throughput (0% packet loss) for real-time traffic inspection.
- Classified network traffic across 15+ applications and protocols and extracted 18+ domain names using TLS SNI analysis, providing visibility into encrypted HTTPS traffic and protocol-level communication.

Publications & Achievements

- **Research Publication:** Multimodal Deep Learning Based Precision Treatment for Lung Cancer (IEEE Xplore, 2026).
- **Generative AI with Claude** (Claude 101, Claude Code 101, Claude Code in Action, Claude Cowork) – Anthropic (2026)
- **AI Fluency: Framework & Foundations** – Anthropic (2026)
- **Machine Learning Master Certification** – Altair RapidMiner (2024)
- **Programming Test Series (C & Python)** and **Programming in Java** – Sanfoundry (2025)
- **Smart India Hackathon 2024** – Advanced to the second round of the Ministry of Education's national-level hackathon.